

IN3140 — Obligatorisk oppgave 2

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Task 1

- 1a)** see IN3140_task1a_fridatbe.py
- 1b)** see IN3140_task1b_fridatbe.py
- 1c)** see IN3140_task1c_fridatbe.py
- 1d)** see IN3140_task1d_fridatbe.py

Task 2

2a)

Because there are three links, the Jacobian will be a 6x3-matrix, and since all three joints are revolute the Jacobian will be on the form:

$$J = \begin{bmatrix} Jv_1 & Jv_2 & Jv_3 \\ Jw_1 & Jw_2 & Jw_3 \end{bmatrix} = \begin{bmatrix} z_0 \times (0_3 - 0_0) & z_1 \times (0_3 - 0_1) & z_2 \times (0_3 - 0_2) \\ z_0 & z_1 & z_2 \end{bmatrix}$$

To derivate the Jacobian we have the calculations of the Denavit Hartenberg matrixes from mandatory assignment 1:

$$H_1^0 H_2^1 H_3^2 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_3 & -s_3 & 0 & a_3 c_3 \\ s_3 & c_3 & 0 & a_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(H_1^0 H_2^1) H_3^2 = H_2^0 H_3^2 = \begin{bmatrix} c_1 c_2 & -s_2 c_1 & -s_1 & a_2 c_1 c_2 \\ s_1 c_2 & -s_1 s_2 & c_1 & a_2 c_2 s_1 \\ -s_2 & -c_2 & 0 & -a_2 s_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_3 & -s_3 & 0 & a_3 c_3 \\ s_3 & c_3 & 0 & a_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_3^0 = \begin{bmatrix} c_1 c_2 c_3 - c_1 s_2 s_3 & -c_1 c_2 s_3 - c_1 s_2 c_3 & -s_1 & a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2 \\ s_1 c_2 c_3 - s_1 s_2 s_3 & -s_1 c_2 s_3 - s_1 s_2 c_3 & c_1 & a_3 s_1 c_2 c_3 - a_3 s_1 s_2 s_3 + a_2 s_1 c_2 \\ -s_2 c_3 - c_2 s_3 & s_2 s_3 - c_2 c_3 & 0 & -a_3 s_2 c_3 - a_3 c_2 s_3 - a_2 s_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

From the matrixes we can read the values of (z_0 and 0_0 is always the same):

$$Jw_1 = z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad Jw_2 = z_1 = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \quad Jw_3 = z_2 = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix}$$

$$0_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad 0_1 = \begin{bmatrix} 0 \\ 0 \\ d_1 \end{bmatrix} \quad 0_2 = \begin{bmatrix} a_2 c_1 c_2 \\ a_2 s_1 c_2 \\ -a_2 s_2 + d_1 \end{bmatrix}$$

$$0_3 = \begin{bmatrix} a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2 \\ a_3 s_1 c_2 c_3 - a_3 s_1 s_2 s_3 + a_2 s_1 c_2 \\ -a_3 s_2 c_3 - a_3 c_2 s_3 - a_2 s_2 + d_1 \end{bmatrix} = \begin{bmatrix} a_3 c_1 c_2 c_3 + a_2 c_1 c_2 \\ a_3 s_1 c_2 c_3 + a_2 s_1 c_2 \\ -a_3 s_2 c_3 - a_2 s_2 + d_1 \end{bmatrix}$$

Calculating the values of the jacobian:

$$Jv_1 = z_0 \times (0_3 - 0_0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \left(\begin{bmatrix} a_3 c_1 c_{23} + a_2 c_1 c_2 \\ a_3 s_1 c_{23} + a_2 s_1 c_2 \\ -a_3 s_{23} - a_2 s_2 + d_1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right)$$

$$Jv_1 = \underline{\begin{bmatrix} -a_3 s_1 c_{23} - a_2 s_1 c_2 \\ a_3 c_1 c_{23} + a_2 c_1 c_2 \\ 0 \end{bmatrix}}$$

$$\begin{aligned} Jv_2 &= z_1 \times (0_3 - 0_1) = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \times \left(\begin{bmatrix} a_3 c_1 c_{23} + a_2 c_1 c_2 \\ a_3 s_1 c_{23} + a_2 s_1 c_2 \\ -a_3 s_{23} - a_2 s_2 + d_1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ d_1 \end{bmatrix} \right) \\ &= \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \times \begin{bmatrix} a_3 c_1 c_{23} + a_2 c_1 c_2 \\ a_3 s_1 c_{23} + a_2 s_1 c_2 \\ -a_3 s_{23} - a_2 s_2 \end{bmatrix} = \begin{bmatrix} c_1(-a_3 s_{23} - a_2 s_2) \\ s_1(-a_3 s_{23} - a_2 s_2) \\ -s_1(a_3 s_1 c_{23} + a_2 s_1 c_2) - c_1(a_3 c_1 c_{23} + a_2 c_1 c_2) \end{bmatrix} \\ &= \begin{bmatrix} -a_3 c_1 s_{23} - a_2 c_1 s_2 \\ -a_3 s_1 s_{23} - a_2 s_1 s_2 \\ -a_3 s_1^2 c_{23} - a_2 s_1^2 c_2 - a_3 c_1^2 c_{23} - a_2 c_1^2 c_2 \end{bmatrix} = \begin{bmatrix} -a_3 c_1 s_{23} - a_2 c_1 s_2 \\ -a_3 s_1 s_{23} - a_2 s_1 s_2 \\ s_1^2(-a_3 c_{23} - a_2 c_2) + c_1^2(-a_3 c_{23} - a_2 c_2) \end{bmatrix} \\ Jv_2 &= \begin{bmatrix} -a_3 c_1 s_{23} - a_2 c_1 s_2 \\ -a_3 s_1 s_{23} - a_2 s_1 s_2 \\ (c_1^2 + s_1^2)(-a_3 c_{23} - a_2 c_2) \end{bmatrix} = \underline{\begin{bmatrix} -a_3 c_1 s_{23} - a_2 c_1 s_2 \\ -a_3 s_1 s_{23} - a_2 s_1 s_2 \\ -a_3 c_{23} - a_2 c_2 \end{bmatrix}} \end{aligned}$$

$$\begin{aligned} Jv_3 &= z_2 \times (0_3 - 0_2) = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \times \left(\begin{bmatrix} a_3 c_1 c_{23} + a_2 c_1 c_2 \\ a_3 s_1 c_{23} + a_2 s_1 c_2 \\ -a_3 s_{23} - a_2 s_2 + d_1 \end{bmatrix} - \begin{bmatrix} a_2 c_1 c_2 \\ a_2 s_1 c_2 \\ -a_2 s_2 + d_1 \end{bmatrix} \right) \\ &= \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \times \begin{bmatrix} a_3 c_1 c_{23} \\ a_3 s_1 c_{23} \\ -a_3 s_{23} \end{bmatrix} = \begin{bmatrix} c_1(-a_3 s_{23}) \\ s_1(-a_3 s_{23}) \\ -s_1(a_3 s_1 c_{23}) - c_1(a_3 c_1 c_{23}) \end{bmatrix} = \begin{bmatrix} -a_3 c_1 s_{23} \\ -a_3 s_1 s_{23} \\ -a_3 s_1^2 c_{23} - a_3 c_1^2 c_{23} \end{bmatrix} \\ Jv_3 &= \begin{bmatrix} -a_3 c_1 s_{23} \\ -a_3 s_1 s_{23} \\ s_1^2(-a_3 c_{23}) + c_1^2(-a_3 c_{23}) \end{bmatrix} = \underline{\begin{bmatrix} -a_3 c_1 s_{23} \\ -a_3 s_1 s_{23} \\ -a_3 c_{23} \end{bmatrix}} \end{aligned}$$

$$J = \underline{\begin{bmatrix} -a_3 s_1 c_{23} - a_2 s_1 c_2 & -a_3 c_1 s_{23} - a_2 c_1 s_2 & -a_3 c_1 s_{23} \\ a_3 c_1 c_{23} + a_2 c_1 c_2 & -a_3 s_1 s_{23} - a_2 s_1 s_2 & -a_3 s_1 s_{23} \\ 0 & -a_3 c_{23} - a_2 c_2 & -a_3 c_{23} \\ 0 & -s_1 & -s_1 \\ 0 & c_1 & c_1 \\ 1 & 0 & 0 \end{bmatrix}}$$

2b)

When the rank of $J(q)$ (number of lineary independent columns in J) is lower than the maximum value (number of columns in J) we have a singular configuration. Because then we have lineary dependent columns in the matrix and there would be either zero or infinitely many solutions to the inverse kinematics.

2c)

By solving the expressions be setting the determinant of the Jacobian to zero we will find the singular configurations (we only view the J_v -matix, where we get the velocity in each of the coordinate's directions):

$$\det(J_v) = \det \left(\begin{bmatrix} -a_3 s_1 c_{23} - a_2 s_1 c_2 & -a_3 c_1 s_{23} - a_2 c_1 s_2 & -a_3 c_1 s_{23} \\ a_3 c_1 c_{23} + a_2 c_1 c_2 & -a_3 s_1 s_{23} - a_2 s_1 s_2 & -a_3 s_1 s_{23} \\ 0 & -a_3 c_{23} - a_2 c_2 & -a_3 c_{23} \end{bmatrix} \right) = 0$$

Part A:

$$-a_3 s_1 c_{23} - a_2 s_1 c_2 \begin{vmatrix} -a_3 s_1 s_{23} - a_2 s_1 s_2 & -a_3 s_1 s_{23} \\ -a_3 c_{23} - a_2 c_2 & -a_3 c_{23} \end{vmatrix}$$

Part B:

$$a_3 c_1 c_{23} + a_2 c_1 c_2 \begin{vmatrix} -a_3 c_1 s_{23} - a_2 c_1 s_2 & -a_3 c_1 s_{23} \\ -a_3 c_{23} - a_2 c_2 & -a_3 c_{23} \end{vmatrix}$$

Part C:

$$0 \begin{vmatrix} -a_3 c_1 s_{23} - a_2 c_1 s_2 & -a_3 c_1 s_{23} \\ -a_3 s_1 s_{23} - a_2 s_1 s_2 & -a_3 s_1 s_{23} \end{vmatrix} = 0$$

Since part C equals 0, all we have to find is the values for when part A equals part B, because then $\det(J_v) = A-B = 0$:

Starting with part A:

$$\begin{aligned} & -a_3 s_1 c_{23} - a_2 s_1 c_2 [(-a_3 s_1 s_{23} - a_2 s_1 s_2)(-a_3 c_{23}) - (-a_3 s_1 s_{23})(-a_3 c_{23} - a_2 c_2)] \\ & -a_3 s_1 c_{23} - a_2 s_1 c_2 [(a_3^2 c_{23} s_1 s_{23} + a_2 a_3 c_{23} s_1 s_2) - (a_3^2 c_{23} s_1 s_{23} + a_2 a_3 c_2 s_1 s_{23})] \\ & -a_3 s_1 c_{23} - a_2 s_1 c_2 [(\cancel{a_3^2 c_{23} s_1 s_{23}} + a_2 a_3 c_{23} s_1 s_2) - (\cancel{a_3^2 c_{23} s_1 s_{23}} + a_2 a_3 c_2 s_1 s_{23})] \\ & \quad -a_3 s_1 c_{23} - a_2 s_1 c_2 [a_2 a_3 c_{23} s_1 s_2 - a_2 a_3 c_2 s_1 s_{23}] \\ & -a_2 a_3^2 c_{23}^2 s_1^2 s_2 + a_2 a_3^2 c_2 c_{23} s_1^2 s_{23} - a_2^2 a_3 c_2 c_{23} s_1^2 s_2 + a_2^2 a_3 c_2^2 s_1^2 s_{23} \\ & -a_2 a_3^2 c_{23}^2 s_1^2 s_2 + a_2 a_3^2 c_2 c_{23} s_1^2 s_{23} - a_2^2 a_3 c_2 c_{23} s_1^2 s_2 + a_2^2 a_3 c_2^2 s_1^2 s_{23} \end{aligned}$$

Then part B:

$$\begin{aligned} & a_3 c_1 c_{23} + a_2 c_1 c_2 [(-a_3 c_1 s_{23} - a_2 c_1 s_2)(-a_3 c_{23}) - (-a_3 c_1 s_{23})(-a_3 c_{23} - a_2 c_2)] \\ & a_3 c_1 c_{23} + a_2 c_1 c_2 [\cancel{a_3^2 c_1 c_{23} s_{23}} + a_2 a_3 c_1 c_{23} s_2 - (\cancel{a_3^2 c_1 c_{23} s_{23}} + a_2 a_3 c_1 c_2 s_{23})] \\ & \quad a_3 c_1 c_{23} + a_2 c_1 c_2 [a_2 a_3 c_1 c_{23} s_2 - a_2 a_3 c_1 c_2 s_{23}] \\ & a_2 a_3^2 c_1^2 c_{23}^2 s_2 - a_2 a_3^2 c_1^2 c_2 c_{23} s_{23} + a_2^2 a_3 c_1^2 c_2 c_{23} s_2 - a_2^2 a_3 c_1^2 c_2^2 s_{23} \\ & a_2 a_3^2 c_1^2 c_{23}^2 s_2 - a_2 a_3^2 c_1^2 c_2 c_{23} s_{23} + a_2^2 a_3 c_1^2 c_2 c_{23} s_2 - a_2^2 a_3 c_1^2 c_2^2 s_{23} \end{aligned}$$

Part A = Part B:

$$\begin{aligned}
& -a_2 a_3^2 c_{23}^2 s_1^2 s_2 + a_2 a_3^2 c_2 c_{23} s_1^2 s_{23} - a_2^2 a_3 c_2 c_{23} s_1^2 s_2 + a_2^2 a_3 c_2^2 s_1^2 s_{23} \\
& = a_2 a_3^2 c_1^2 c_{23}^2 s_2 - a_2 a_3^2 c_1^2 c_2 c_{23} s_{23} + a_2^2 a_3 c_1^2 c_2 c_{23} s_2 - a_2^2 a_3 c_1^2 c_2^2 s_{23} \\
& a_2 a_3^2 c_{23}^2 s_2 (c_1^2 + s_1^2) - a_2 a_3^2 c_2 c_{23} s_{23} (c_1^2 + s_1^2) + a_2^2 a_3 c_2 c_{23} s_2 (c_1^2 + s_1^2) - a_2^2 a_3 c_2^2 s_{23} (c_1^2 + s_1^2) = 0 \\
& a_2 a_3^2 c_{23}^2 s_2 - a_2 a_3^2 c_2 c_{23} s_{23} + a_2^2 a_3 c_2 c_{23} s_2 - a_2^2 a_3 c_2^2 s_{23} = 0 \\
& a_2 a_3^2 c_{23} (s_2 c_{23} - s_{23} c_2) + a_2^2 a_3 c_2 (c_{23} s_2 - s_{23} c_2) = 0 \\
& a_2 a_3^2 c_{23} (s_2 (c_2 c_3 - s_2 s_3) - c_2 (s_2 c_3 + c_2 s_3)) + a_2^2 a_3 c_2 (s_2 (c_2 c_3 - s_2 s_3) - c_2 (s_2 c_3 + c_2 s_3)) = 0 \\
& a_2 a_3^2 c_{23} (\cancel{c_2 c_3 s_2} - s_2^2 s_3 - \cancel{c_2 c_3 s_2} - c_2^2 s_3) + a_2^2 a_3 c_2 (\cancel{c_2 c_3 s_2} - s_2^2 s_3 - \cancel{c_2 c_3 s_2} - c_2^2 s_3) = 0 \\
& a_2 a_3^2 c_{23} (-s_2^2 s_3 - c_2^2 s_3) + a_2^2 a_3 c_2 (-s_2^2 s_3 - c_2^2 s_3) = 0 \\
& a_2 a_3^2 c_{23} (-s_3 (s_2^2 + c_2^2)) + a_2^2 a_3 c_2 (-s_3 (s_2^2 + c_2^2)) = 0 \\
& -s_3 (a_2 a_3^2 c_{23} + a_2^2 a_3 c_2) = 0 \\
& -s_3 = 0 \quad \vee \quad a_2 a_3 (a_3 c_{23} + a_2 c_2) = 0
\end{aligned}$$

We know that a_2 and a_3 are 222.1 and 136.2, thus not zero:

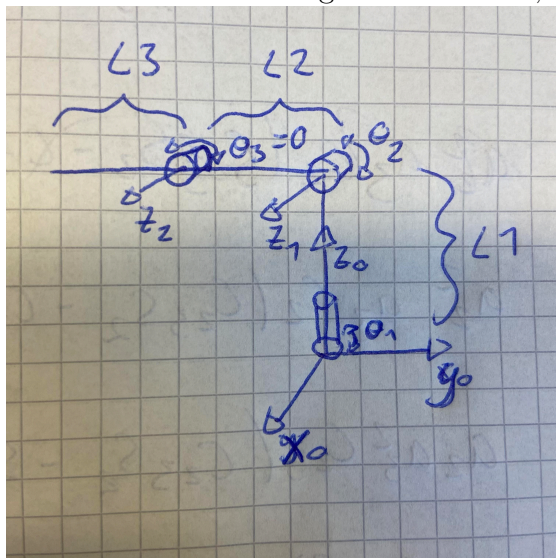
$$\begin{aligned}
a_2 a_3 (a_3 c_{23} + a_2 c_2) &= 0 \\
a_3 c_{23} + a_2 c_2 &= 0 \\
\underline{a_3 (c_2 c_3 - s_2 s_3) + a_2 c_2} &= 0
\end{aligned}$$

And we know the value of θ_3 when $\sin(\theta) = 0$:

$$s_3 = 0 \quad \rightarrow \quad \theta_3 = \underline{0} \quad \vee \quad \theta_3 = \underline{\pi}$$

2d)

I chose to show the singularity when $\theta_3 = 0$, which makes link 2 and 3 collinear, and it would be like link 2 and 3 is one combined link. This makes the robot have one less degree of freedom, and therefore is defined as a singularity.

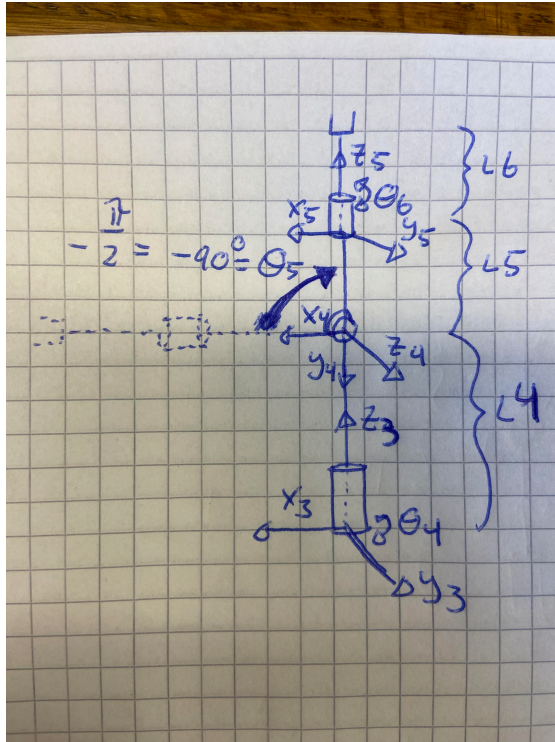


Viewing the two other results:

- $\theta_3 = \pi$, because the tip and link 3 will lay directly on top of link 2 (one less degree of freedom)
- $a_3(c_2c_3 - s_2s_3) + a_2c_2 = 0$, because in this case the tip will intersect the z_0 -axis, and there would be infinitely many solutions to the inverse kinematics because θ_1 does not matter

2e)

Example of a singularity for the spherical wrist:



This is a singularity because the z_5 -axis and z_3 -axis are collinear. If $\theta_5 = \frac{\pi}{2}$ another singularity would appear. In these cases, there would be infinitely many solutions to the inverse kinematics because the angle of θ_4 and θ_6 would not matter for the point of the gripper.

When $\theta_5 = -90^\circ$ the wrist is limited to one less degree of freedom because it would be like link 4 and link 5 is one combined link. Also joint 4 and 6 would have the same effect on the wrist (unless one or both of them were limited to less than 360 degrees of rotation), so this would make the gripper to have the same degree of freedom as a one-revoloute-joint wrist ($\text{DOF} = 1$) instead of three. And the same situations will appear when $\theta_5 = 90^\circ$, because link 6 would be placed on top of link 4 (be collinear)

2f)

When the robot ends up in a singular configuration, it cannot move in some direction/directions. This means that if the robot ends up in a singular configuration, it (the robot) might get destroyed if it tries to move in the 'impossible' direction because it would theoretically require infinite force to get out of the position.

Task 3

3a) see IN3140_task3a_fridatbe.py

3b) see IN3140_task3b_fridatbe.py